

```
In [6]: l1=[]  
print(type(l1))
```

```
<class 'list'>
```

```
In [7]: print(type(l1))
```

```
<class 'list'>
```

```
In [8]: l2=[23,54,67,223,295] #only integer values  
l2
```

```
Out[8]: [23, 54, 67, 223, 295]
```

```
In [9]: l3=[23.4, 95.6, 4390.5] #only float values  
l4=['yokoso', 'watashi no', 'soul society ye']  
l4
```

```
Out[9]: ['yokoso', 'watashi no', 'soul society ye']
```

```
In [12]: l5=['bankai', [12,43,61], [56,94,74], 91] #list with nested lists  
l6=[29, 50.51, 'titan'] #mixed values  
print(l5)  
print(l6)
```

```
['bankai', [12, 43, 61], [56, 94, 74], 91]  
[29, 50.51, 'titan']
```

```
In [13]: len(l6)
```

```
Out[13]: 3
```

```
In [14]: len(l5)
```

```
Out[14]: 4
```

```
In [15]: l2[0] #forward indexing
```

```
Out[15]: 23
```

```
In [16]: l5[3]
```

```
Out[16]: 91
```

```
In [17]: l4[2]
```

```
Out[17]: 'soul society ye'
```

```
In [18]: l4[-2] #backward indexing
```

```
Out[18]: 'watashi no'
```

List Slicing

```
In [21]: mylist = ['wake', 'to', 'reality!', 'nothing', 'goes', 'as', 'planned', 'in', 't
```

```
In [20]: mylist[:]
```

```
Out[20]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [22]: mylist[0:3]
```

```
Out[22]: ['wake', 'to', 'reality!']
```

```
In [23]: mylist[2:5]
```

```
Out[23]: ['reality!', 'nothing', 'goes']
```

```
In [24]: mylist[:3]
```

```
Out[24]: ['wake', 'to', 'reality!']
```

```
In [25]: mylist[3:]
```

```
Out[25]: ['nothing', 'goes', 'as', 'planned', 'in', 'this', 'accursed', 'world']
```

```
In [26]: mylist[-1:]
```

```
Out[26]: ['world']
```

```
In [27]: mylist[:-1]
```

```
Out[27]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed']
```

```
In [30]: mylist[-9:]
```

```
Out[30]: ['reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [31]: len(mylist)
```

```
Out[31]: 11
```

```
In [32]: mylist.append('!!') #adds a new value at the end of the list
```

```
In [33]: mylist
```

```
Out[33]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world',  
          '!!']
```

```
In [36]: mylist.append('the longer you live, the more you realize that the only things th  
mylist
```

```
Out[36]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world',  
          '!!',  
          'the longer',  
          'the longer you live, the more you realize that the only things that truly exi  
st in this reality are merely pain',  
          'the longer you live, the more you realize that the only things that truly exi  
st in this reality are merely pain']
```

```
In [37]: mylist.pop(13) #removes the values from the mentioned index values
```

```
Out[37]: 'the longer you live, the more you realize that the only things that truly exis  
t in this reality are merely pain'
```

```
In [38]: mylist.pop(12)
```

```
Out[38]: 'the longer'
```

```
In [39]: mylist
```

```
Out[39]: ['wake',
          'to',
          'reality!',
          'nothing',
          'goes',
          'as',
          'planned',
          'in',
          'this',
          'accursed',
          'world',
          '!',
          'the longer you live, the more you realize that the only things that truly exist in this reality are merely pain']
```

```
In [40]: len(mylist)
```

```
Out[40]: 13
```

```
In [41]: mylist.insert(14, 'suffering and futility') #inserts value in the specifies index
```

```
In [42]: mylist
```

```
Out[42]: ['wake',
          'to',
          'reality!',
          'nothing',
          'goes',
          'as',
          'planned',
          'in',
          'this',
          'accursed',
          'world',
          '!',
          'the longer you live, the more you realize that the only things that truly exist in this reality are merely pain',
          'suffering and futility']
```

```
In [43]: mylist.remove('suffering and futility') #removed the mentioned value
```

```
In [44]: mylist
```

```
Out[44]: ['wake',
          'to',
          'reality!',
          'nothing',
          'goes',
          'as',
          'planned',
          'in',
          'this',
          'accursed',
          'world',
          '!',
          'the longer you live, the more you realize that the only things that truly exist in this reality are merely pain']
```

```
In [45]: mylist.remove('the longer you live, the more you realize that the only things th
```

```
In [46]: mylist
```

```
Out[46]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world',  
          '!']
```

```
In [47]: mylist.clear #'clear()' will clear the entire list
```

```
Out[47]: <function list.clear()>
```

```
In [48]: mylist.clear()
```

```
In [49]: mylist
```

```
Out[49]: []
```

```
In [50]: mylist = ['wake', 'to', 'reality!', 'nothing', 'ever', 'goes', 'as', 'planned',  
mylist
```

```
Out[50]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'ever',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [54]: mylist1 = mylist.copy() #'copy()' will copy the values of one list to another
```

```
In [57]: mylist
```

```
Out[57]: []
```

```
In [58]: mylist = ['wake', 'to', 'reality!', 'nothing', 'ever', 'goes', 'as', 'planned',  
mylist
```

```
Out[58]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'ever',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [59]: mylist1=mylist.copy()
```

```
In [60]: mylist1
```

```
Out[60]: ['wake',  
          'to',  
          'reality!',  
          'nothing',  
          'ever',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [62]: id(mylist)
```

```
Out[62]: 2491890391680
```

```
In [64]: id(mylist1)
```

```
Out[64]: 2491890833856
```

```
In [66]: mylist==mylist1
```

```
Out[66]: True
```

```
In [67]: mylist1[1]='madara'
```

```
In [68]: mylist1
```

```
Out[68]: ['wake',  
          'madara',  
          'reality!',  
          'nothing',  
          'ever',  
          'goes',  
          'as',  
          'planned',  
          'in',  
          'this',  
          'accursed',  
          'world']
```

```
In [69]: list1 = ['time', 'doesnt', 'heal', 'anything', ',', ',']  
list2 = ['it', 'teaches', 'us', 'how', 'to', 'live', 'with', 'it', '.']  
print(list1, list2)
```

```
['time', 'doesnt', 'heal', 'anything', ',', ','] ['it', 'teaches', 'us', 'how', 'to', 'live', 'with', 'it', '.']
```

```
In [70]: list1.extend(list2) # 'extend' will merge 2 lists  
list1
```

```
Out[70]: ['time',  
          'doesnt',  
          'heal',  
          'anything',  
          ',',  
          ',',  
          'it',  
          'teaches',  
          'us',  
          'how',  
          'to',  
          'live',  
          'with',  
          'it',  
          '.']
```

```
In [71]: list1
```

```
Out[71]: ['time',  
          'doesnt',  
          'heal',  
          'anything',  
          ',',  
          ',',  
          'it',  
          'teaches',  
          'us',  
          'how',  
          'to',  
          'live',  
          'with',  
          'it',  
          '.']
```

```
In [72]: 'with' in list1
```

```
Out[72]: True
```

```
In [75]: 'holy' in list1
```

```
Out[75]: False
```

```
In [73]: if 'with' in list1:  
          print('Itachi')  
else:  
          print('Notg itachi')
```

```
Itachi
```

```
In [74]: if 'madara' in list1:  
          print('itachi')
```

```
else:  
    print('Madara')
```

Madara

In [76]: lis1

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[76], line 1  
----> 1 lis1  
  
NameError: name 'lis1' is not defined
```

In [77]: list

Out[77]: list

In [78]: list1

Out[78]: ['time',
 'doesn't',
 'heal',
 'anything',
 ',',
 'it',
 'teaches',
 'us',
 'how',
 'to',
 'live',
 'with',
 'it',
 '.']

In [81]: list1.reverse()
list1

Out[81]: [',',
 '.',
 'anything',
 'doesn't',
 'heal',
 'how',
 'it',
 'it',
 'live',
 'teaches',
 'time',
 'to',
 'us',
 'with']

In [85]: list1=list1[::-1]
list1


```
Out[85]: ['with',  
          'us',  
          'to',  
          'time',  
          'teaches',  
          'live',  
          'it',  
          'it',  
          'how',  
          'heal',  
          'doesnt',  
          'anything',  
          '.',  
          ',']
```

```
In [88]: lis3 = [2,53,3,67,6,89,5,9,89,0]  
lis3.sort()  
lis3
```

```
Out[88]: [0, 2, 3, 5, 6, 9, 53, 67, 89, 89]
```

```
In [90]: lis3.sort(reverse=True)  
lis3
```

```
Out[90]: [89, 89, 67, 53, 9, 6, 5, 3, 2, 0]
```

```
In [91]: list1
```

```
Out[91]: ['with',  
          'us',  
          'to',  
          'time',  
          'teaches',  
          'live',  
          'it',  
          'it',  
          'how',  
          'heal',  
          'doesnt',  
          'anything',  
          '.',  
          ',']
```

```
In [92]: for i in list1:  
          print(i)
```

```
with
us
to
time
teaches
live
it
it
how
heal
doesnt
anything
.
,
```

```
In [97]: for i in enumerate(list1):
        print(i)
```

```
(0, 'with')
(1, 'us')
(2, 'to')
(3, 'time')
(4, 'teaches')
(5, 'live')
(6, 'it')
(7, 'it')
(8, 'how')
(9, 'heal')
(10, 'doesnt')
(11, 'anything')
(12, '.')
(13, ',')
```

```
In [96]: list1.count('in')
```

```
Out[96]: 0
```

```
In [101... lis1 = [8,6,1,2]
all(lis1)
```

```
Out[101... True
```

```
In [102... any(lis1)
```

```
Out[102... True
```

```
In [ ]:
```